Midterm Exam I (120 minute exam)

Course: Electromagnetic Theory **Date:** Apr. 04, 2020
Instructor: Dr. Shafayat Abrar **Marks:** 100 Marks

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Instructions:

- Attempt all questions.
- Discussing with each other is not allowed.
- Sharing your work with others is not allowed.

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1. (10 points) A circular doughnut-shaped disc of inner radius a and outer radius b carries a variable surface charge density $\rho_S = c\rho$ [C/m^2] and is placed on the xy-plane with axis the same as the z-axis. Find the electric field intensity $\mathbf{E}$ at $(0, 0, h)$.

   Note in $\rho_S = c\rho$, c is positive constant and ρ is the distance from z-axis.

2. (40 points) The volume charge density inside an atomic nucleus of radius a is

$$\rho_v = \rho_0 \left(\frac{a^3 - r^3}{a^3} \right)$$

   where ρ_0 is a constant.

   [10] **(a)** Calculate the total charge.
   [15] **(b)** Determine $\mathbf{E}$ and V outside the nucleus.
   [15] **(c)** Determine $\mathbf{E}$ and V inside the nucleus.

3. (10 points) Determine the amount of work needed to transfer two charges of 20 nC, 30 nC and -40 nC from infinity to locations $(0, 0, 1)$, $(0, 1, 0)$ and $(1, 0, 0)$, respectively.

4. (20 points) Let $\mathbf{D} = \rho^2 z \mathbf{a}_\rho + \rho \sin^2(\phi) \mathbf{a}_z$. Evaluate

   [10] **(a)** $\displaystyle\oint_S \mathbf{D} \cdot \mathrm{d}\mathbf{S}$
   [10] **(b)** $\displaystyle\int_v \boldsymbol{\nabla} \cdot \mathbf{D} \, \mathrm{d}v$
   over the region defined by $1 < \rho < 5$, $0 < z < 1$, $0 < \phi < 2\pi$.

5. (20 points) If $\mathbf{A} = \rho z^2 \, \mathbf{a}_\rho + z \cos(\phi) \, \mathbf{a}_\phi + \rho^2 \sin(\phi) \, \mathbf{a}_z$ verify Stokes's theorem for the open surface defined by $z = 2$, $0 < \rho < 3$, $0 < \phi < 90°$.